

LESSON 128 (0.5) Ground (1.5) Local Dual

Lesson Objective

The purpose of this lesson is to introduce the applicant to short and soft-field takeoffs and landings. The applicant will gain an understanding of why and how these operations are conducted. This lesson will also introduce to the applicant the concept of steep turns and emergency descents.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- During this lesson quiz the student on the following and provide instruction where needed.
- Student questions.

References

- Airplane Flying handbook (FAA-H-8083-3C)
 - Chapter 6: Takeoffs and Departure Climbs
 - Chapter 9: Approaches and Landings
 - Chapter 10: Performance Maneuvers
 - Chapter 18: Emergency Procedures
- Instrument Flying handbook (FAA-H-8083-15B)
 - Chapter 7: Basic Flight Maneuvers using analog/Electronic Flight Display

Instructor Guidance

- Utilize the tell-demonstrate-practice technique throughout every maneuver in this lesson. Instructor demonstration is critical — do not hand over controls on any new maneuver without a complete demonstration first. Encourage the student to follow along lightly on the controls during your demonstration to build feel and awareness before they attempt it independently.
- Emphasis should be placed on the lesson completion standards rather than ACS standards at this stage. The student is being introduced to these maneuvers for the first time in the aircraft — depth and confidence within the lesson tolerances matter more than checkride precision right now.
- Before departing, quiz the student on the procedures from Lesson 127 homework — short-field, soft-field, steep turns, and emergency descent sequences. If they cannot walk through the procedures from memory, address it on the ground before flying. Students who arrive unprepared for this lesson will absorb significantly less from the demonstrations.
- For short-field takeoffs, verify the student holds the brakes until full static RPM before releasing. A premature brake release results in a longer ground roll and defeats the purpose of the technique. Also verify the student is pitching for a sight picture rather than chasing the airspeed indicator during the V_x climb.
- For soft-field takeoffs, coach the student through staying in ground effect after rotation. The most common error is pitching too aggressively after liftoff and climbing out of ground effect before reaching V_y — resulting in a sluggish, dragging climb. Reinforce full aft stabilator on the roll and forward pressure to stay low after liftoff.

- For short-field landings, emphasize that if the aiming point cannot be made, a go-around is the only correct action. Do not allow the student to force the airplane onto the runway past the intended touchdown point. This habit must be established firmly from the first short-field landing.
- For soft-field landings, coach the student through staying in ground effect during the flare and holding the nosewheel off as long as aerodynamically possible during the rollout. Minimal braking must be reinforced consistently — hard braking on a soft surface is exactly the scenario this technique is designed to avoid.
- Schedule steep turns toward the end of the lesson to minimize the likelihood of motion discomfort for both you and the student. If the student is showing signs of discomfort earlier in the lesson, use judgment about whether to proceed with steep turns or defer them to Lesson 130.
- For steep turns, watch closely for students who apply bank and back pressure too aggressively at entry. A rapid entry at 45° combined with excessive back pressure significantly increases the risk of an accelerated stall — smooth and deliberate inputs must be emphasized from the first demonstration.
- For the emergency descent, after verifying power idle, the student should immediately begin banking and pitching down simultaneously — not sequentially. Students who pause between power reduction and initiating the descent lose valuable altitude. Reinforce the combined simultaneous inputs from the demonstration.

Lesson Content

Checklist Procedures

- Re-emphasize the importance of using checklists.
- Have the student run through all the checklists.
- Student should be able to perform all briefings.

Preflight Inspection

- Ensure that the student conducted a thorough preflight inspection.

Operation of the Communications and Navigation Units

- Ensure that the student understands how to properly use COMS and NAV units.

Engine Starting

- Correctly determine and use appropriate start procedure.
- Incorporate the use of flows.

Radio Communication Procedures and ATC Phraseology

- The student should be able to conduct radio calls in towered airport.
- The students should be able to conduct radio calls in the practice area.

Visual Scanning and Collision Avoidance

- Emphasize that it is the pilot in command's responsibility to avoid traffic.

- IF YOU SEE SOMETHING, SAY SOMETHING!

Taxiing

- The student should be able to taxi on the centerline at appropriate speeds on the ramp and taxiways without instructor guidance
- Ensure the correct use of wind correction throughout taxi.

Short-Field Takeoff and Maximum Performance Climb

- Instructor does, instructor says:
 - Demonstrate a short field takeoff.
 - Talk through runway entry, takeoff roll, and climb out.
 - Emphasize pitching for an attitude and not chasing an airspeed.
 - Use a mix of visual and instrument references.
- Student does, student says:
 - Verify correct procedure.
 - Ensure that the student holds the brakes until full static RPM.
 - Verify that the climb out is conducted at 63 knots.
 - Verify that the student says obstacle cleared if there is an obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.

Soft-Field Takeoff

- Instructor does, instructor says:
 - Demonstrate a soft field takeoff.
 - Talk through runway entry, takeoff roll, and climb out.
 - Emphasize pitching for an attitude and not chasing an airspeed.
 - Use a mix of visual and instrument references.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through maintaining full aft stabilator and staying in ground effect.
 - Verify that the climb out is conducted at 63 or 79 knots depending on obstacles.
 - Verify that the student says obstacle cleared if there is an obstacle.
 - Verify that the student says positive rate of climb before retracting the flaps.

Steep Turns

- Instructor does, student says:
 - Entering the turn:
 - Ensure pre-maneuver check is complete.
 - Choose a visual reference point and bug it in the HSI.
 - Smooth level turn to 45 degrees of bank (too aggressive with the bank and back pressure can result in an accelerated stall!)
 - At 30 degrees of bank, add 100-200 RPM and 3 rolls of aft trim.
 - Maintaining the turn:
 - Focus should primarily be outside.

- Show the student where the horizon intersects with the panel/cowling. The student should maintain that sight picture.
- Coordination.
- Reference the instruments.
- Corrections:
 - If the nose falls below the desired sight picture, pitch needs to increase.
 - If the nose rises above the desired sight picture, pitch needs to decrease.
 - Power should only be used to correct for airspeed changes.
- Rolling out:
 - Roll-out should commence at $\frac{1}{2}$ the bank angle.
 - Pitch the nose down to counteract the decrease in horizontal lift and increase in vertical lift.
 - Bring the power back to cruise power.
 - Trim for level flight.
- Student says, student does:
 - Verify correct procedure.

Forward Slip to a Landing

- Establish the aircraft on final approach with the desired landing configuration and maintain the recommended approach airspeed.
- If the aircraft is too high on the approach, reduce power as needed.
- Apply aileron in the direction of the desired ground track to prevent drifting from the runway centerline.
- Apply opposite rudder to yaw the aircraft and create the slip. This cross-controlled input increases drag and allows the aircraft to descend more rapidly without increasing airspeed.
- Maintain proper approach airspeed with pitch control while adjusting the amount of slip as needed to control the descent rate.
- Continuously monitor runway alignment and airspeed while maintaining the slip.
- Gradually remove the cross-control inputs as the aircraft approaches the flare.
- Align the aircraft with the runway centerline and complete a normal landing flare and touchdown.

Go-Arounds or Rejected Landings

- Have your student practice go-around procedures.
- Encourage your student to call for a go-around themselves rather than waiting for you to call it.
 - If you do call for a go-around, the student should immediately execute the procedure rather than discussing it first.

Short-Field Approach and Landing

- Instructor does, instructor says:
 - Demonstrate a short field landing.
 - Pick a touchdown point and aiming point on the downwind.
 - Talk through the difference between a normal approach and a short field approach

- Emphasize trimming!
- Talk through maintaining a steeper rate of descent and focusing on the aiming point.
- Talk through the round out and flare.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through staying higher on the approach and slower.
 - Coach the student through maintaining a steeper rate of descent and focusing on the aiming point.
 - Help on the round out and flare if needed.
 - Emphasize the importance of not forcing the airplane on the ground. If the point cannot be made, perform a go-around.
 - Verify "maximum" braking.

Soft-Field Approach and Landing

- Instructor does, instructor says:
 - Demonstrate a soft field landing.
 - A normal approach must be conducted.
 - Talk through the round out, staying in ground effect, and timing the flare.
- Student does, student says:
 - Verify correct procedure.
 - Coach the student through the round out, staying in ground effect, and flaring.
 - Verify minimal braking.

Emergency Descent

- Instructor does, instructor says:
 - Demonstrate an emergency descent.
 - Emphasize adding flaps while simultaneously banking and pitching down.
 - Emphasize the importance of looking outside and scanning for traffic throughout the maneuver.
- Student does, student says:
 - Verify the correct procedure.
 - After verifying power idle, coach the student to immediately begin the bank and pitch down, followed by adding flaps.
 - Help the student maintain the nose-down attitude and avoid slowing down too early.
 - Emphasize smoothly applying back pressure on the yoke during recovery.
 - Retract flaps incrementally during the recovery.

Postflight Procedures

- Ensure that the student conducts a thorough postflight inspection.

Post-Brief

- Student self-critique.
- Lesson grading.

- With Primary instructor
- Logbook completion.
 - Description of activities input by student.
- Schedule the next lesson.
- Preview the next lesson → 124.
- Assign homework:
 - Read chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Read chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
 - Read chapter 05 of the Pilot's Handbook of Aeronautical Knowledge - Aerodynamics of Flight
 - Wingtip Vortices.

Completion Standards

This lesson is complete when the applicant can:

1. Demonstrate a basic knowledge and understanding of short-field and soft-field takeoffs and landings.
2. Execute short and soft-field takeoffs and landings with instructor guidance.
3. Conduct all normal and emergency checklist procedures without instructor guidance.
4. Conduct steep turns with instructor guidance and maintain altitude ± 150 ft, airspeed ± 15 knots, bank angle $\pm 10^\circ$ and roll out on the entry heading ± 15 degrees.
5. Conduct emergency descents while maintaining airspeed $+0/-10$ knots, using a bank angle between 30° and 45° , and levelling off at a specified altitude ± 100 ft with instructor guidance.